

Interacting Multipoles of Rank 1 and 3:

$$T_{\alpha\beta\gamma\delta}\mu_{\alpha}\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha} \cdot \Omega_{\beta\gamma\delta} \end{aligned} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{aligned} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} + 3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (2b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} + 6 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (2c)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{zzzz} - 15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} - 15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} - 15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} + 3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} - 15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} + 3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (3c)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \end{array} \right] \quad (4)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot [-45 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz}] \quad (5)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = [-3 \cdot R_z^{-5} \cdot \mu_y \cdot \Omega_{yzz}] \quad (6)$$